



NEW YORK UNIVERSITY

ConvNets in Practice

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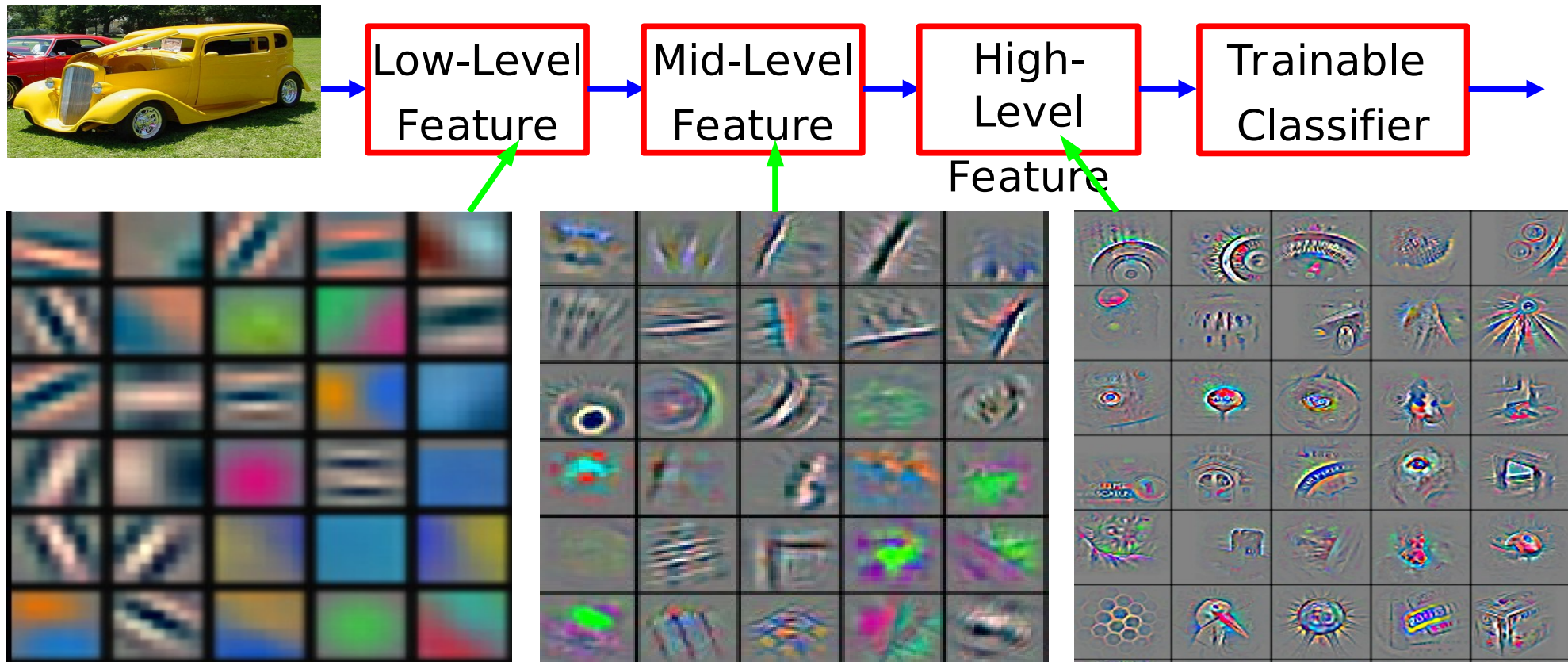
Deep Learning, NYU Spring 2021

What are ConvNets Good For

- Signals that comes to you in the form of (multidimensional) arrays.
- Signals that have strong **local correlations**
- Signals where features can **appear anywhere**
- Signals in which objects are subject to translations and distortions.
- **1D ConvNets: sequential signals, text**
 - Text, music, audio, speech, time series.
- **2D ConvNets: images, time-frequency representations (speech and audio)**
 - Object detection, localization, recognition
- **3D ConvNets: video, volumetric images, tomography images**
 - Video recognition / understanding
 - Biomedical image analysis
 - Hyperspectral image analysis

Deep Learning = Learning Hierarchical Representations

It's **deep** if it has **more than one stage** of non-linear feature transformation

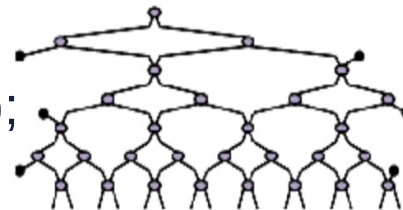
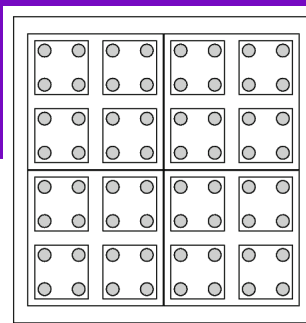


Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

The world is compositional

Convolutional networks learn hierarchical representations

- ▶ Upper-layer representations are at a coarse spatial scale
- ▶ Compositional hierarchies In physics: Renormalization group theory; Multi-scale entanglement renormalization ansatz (MERA);

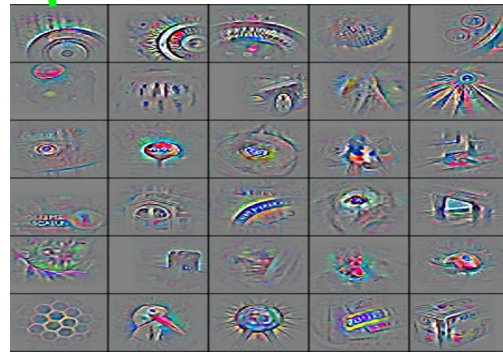
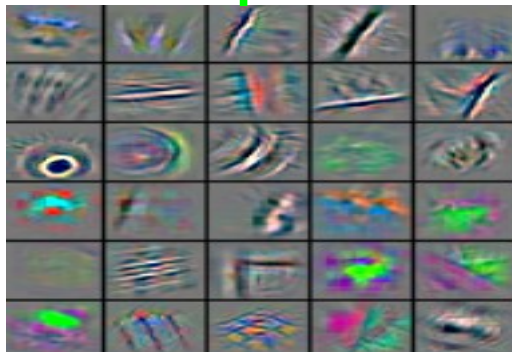


Low-Level
Features

Mid-Level
Features

High-Level
Features

Trainable
Classifier



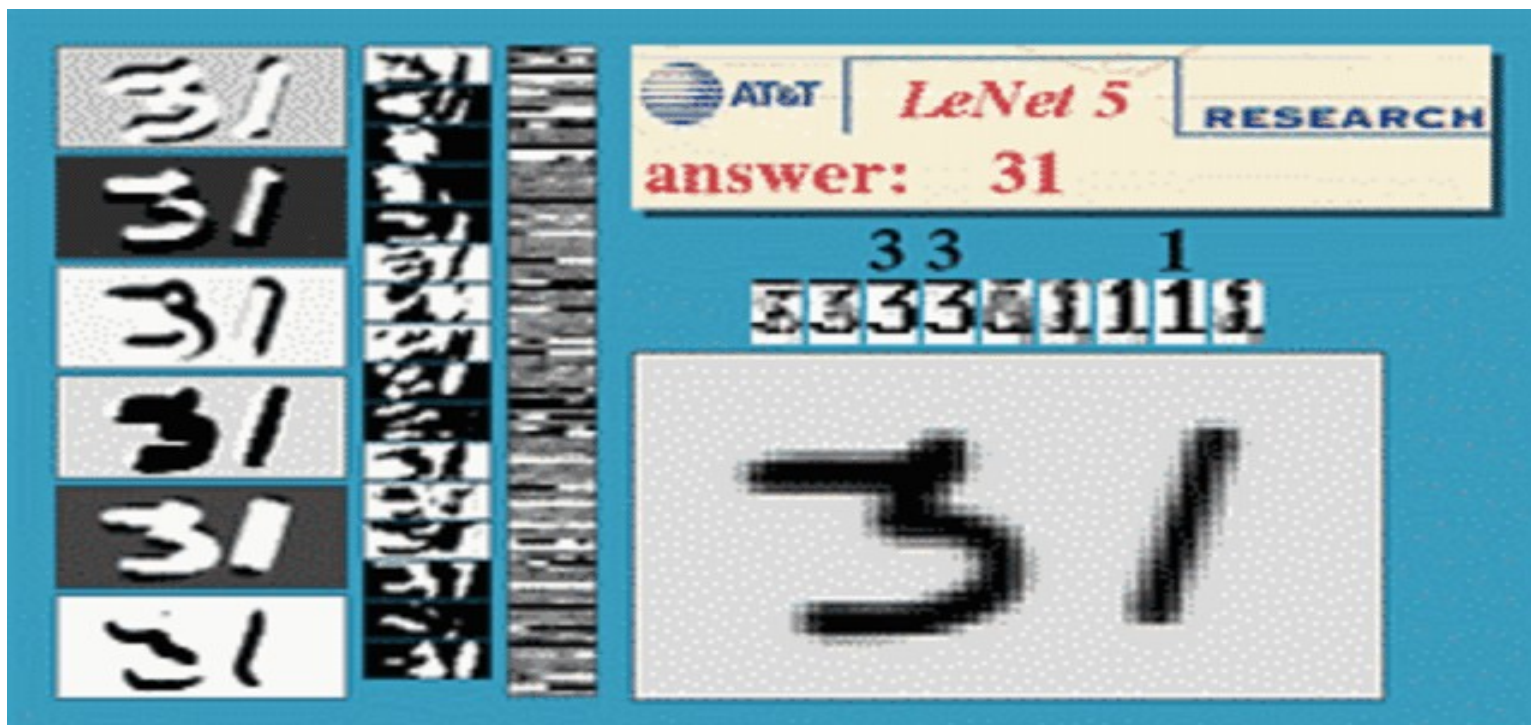
Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Object Detection, Multiple Object Recognition

- Variable-size input.
- Every layer is a convolution, even what is often call “fully-connected layers”.

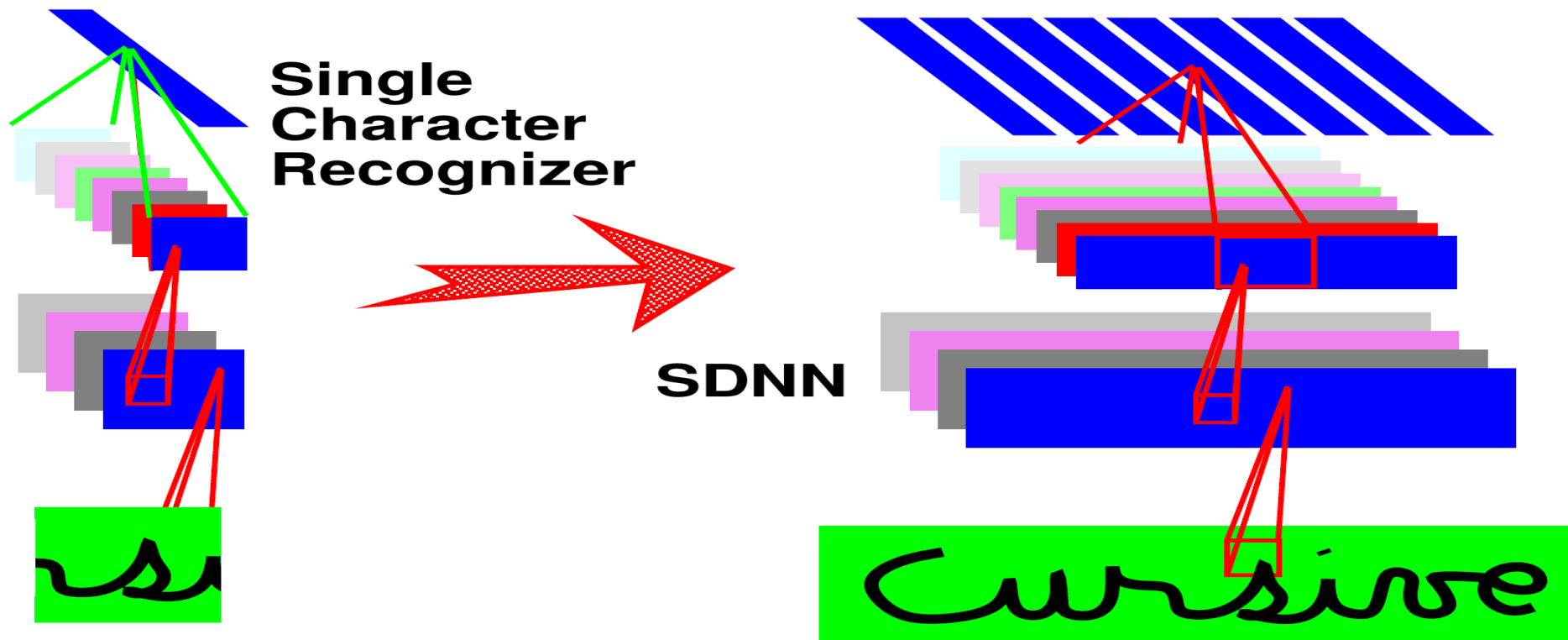


Sliding Window ConvNet + Weighted FSM

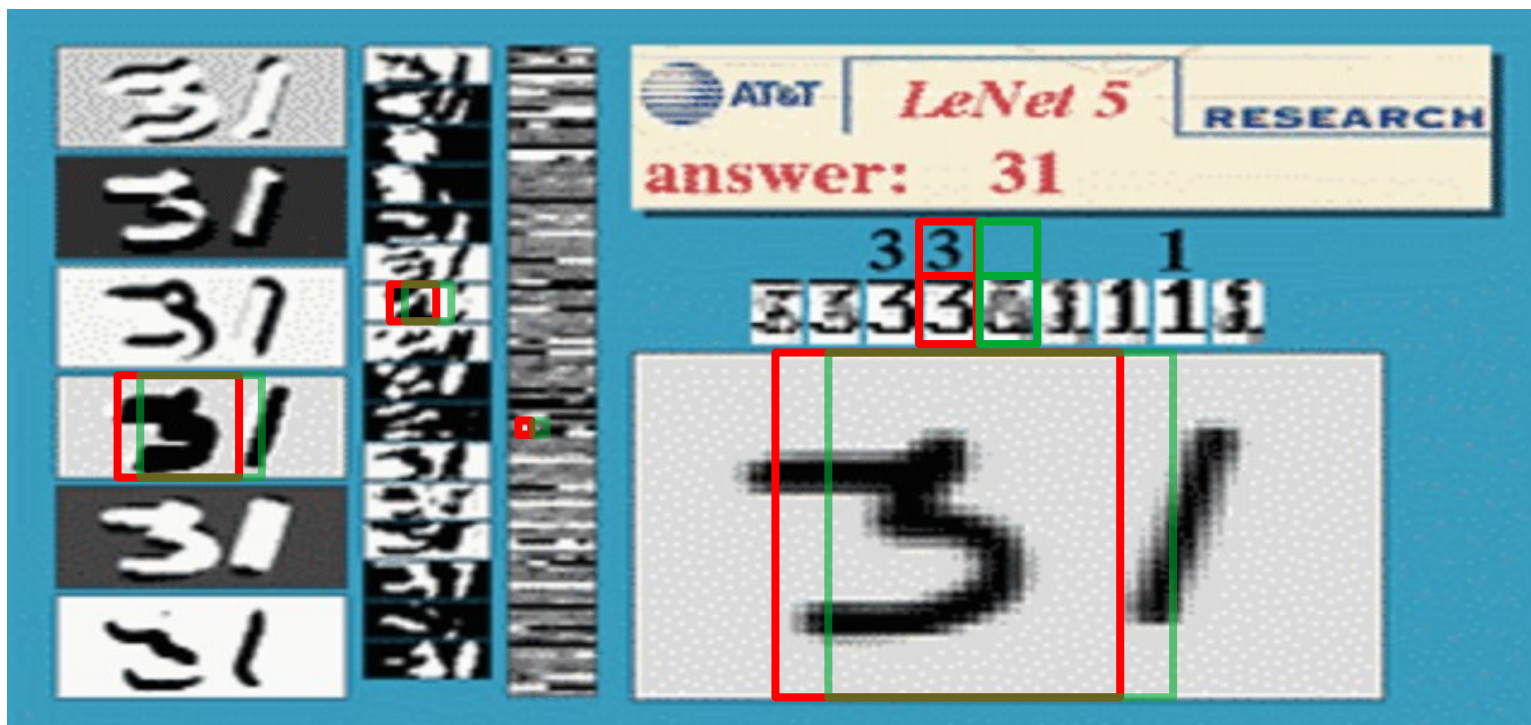


Multiple Character Recognition [Matan et al 1992]

■ Every layer is a convolution

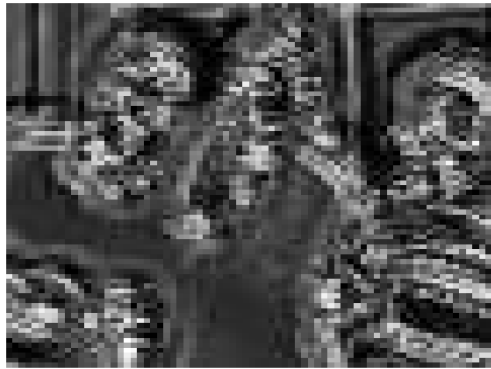
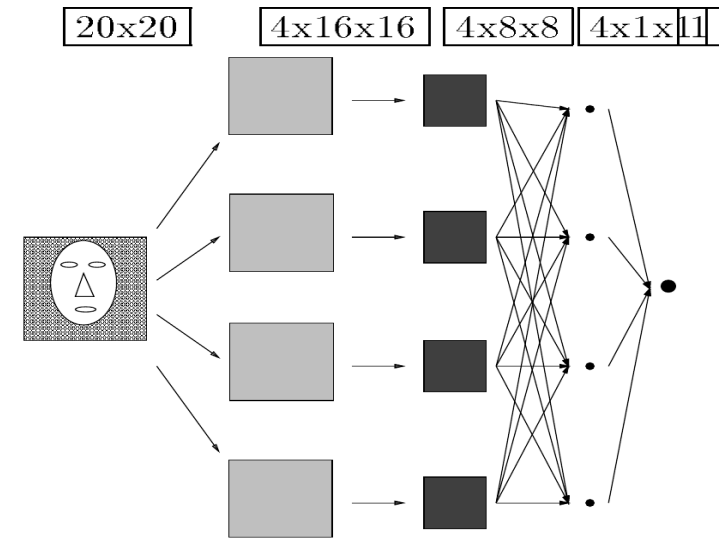


Sliding Window ConvNet + Weighted FSM

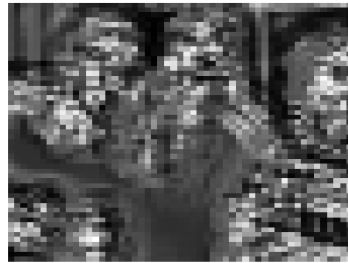


Face Detection [Vaillant et al. 93, 94]

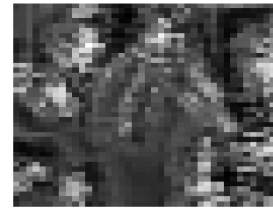
- ConvNet applied to large images
- Heatmaps at multiple scales
- Non-Maximum Suppression for candidates
- 6 second on a Sparcstation for 256x256



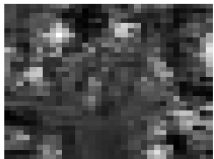
Scale 3



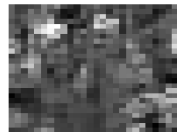
Scale 4



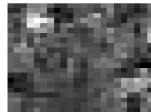
Scale 5



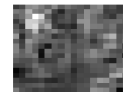
Scale 6



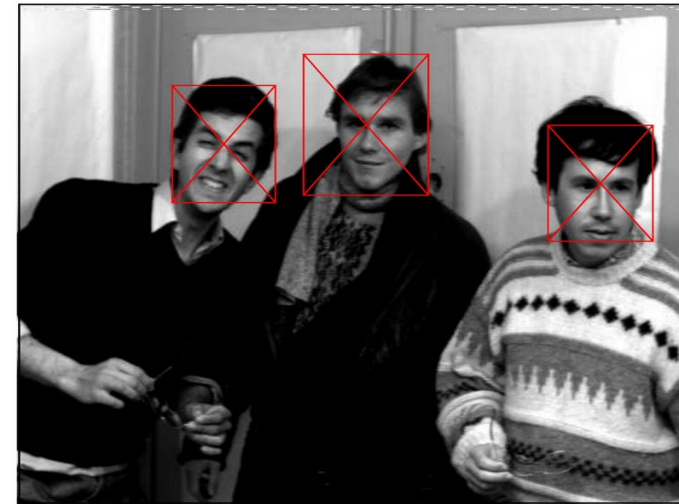
Scale 7



Scale 8



Scale 9





Simultaneous face detection and pose estimation



[Osadchy, Miller, LeCun 2004, 2007]

Image Segmentation with ConvNets

Biological Image Segmentation

► [Ning et al. IEEE-TIP 2005]

Pixel labeling with large context using a convnet

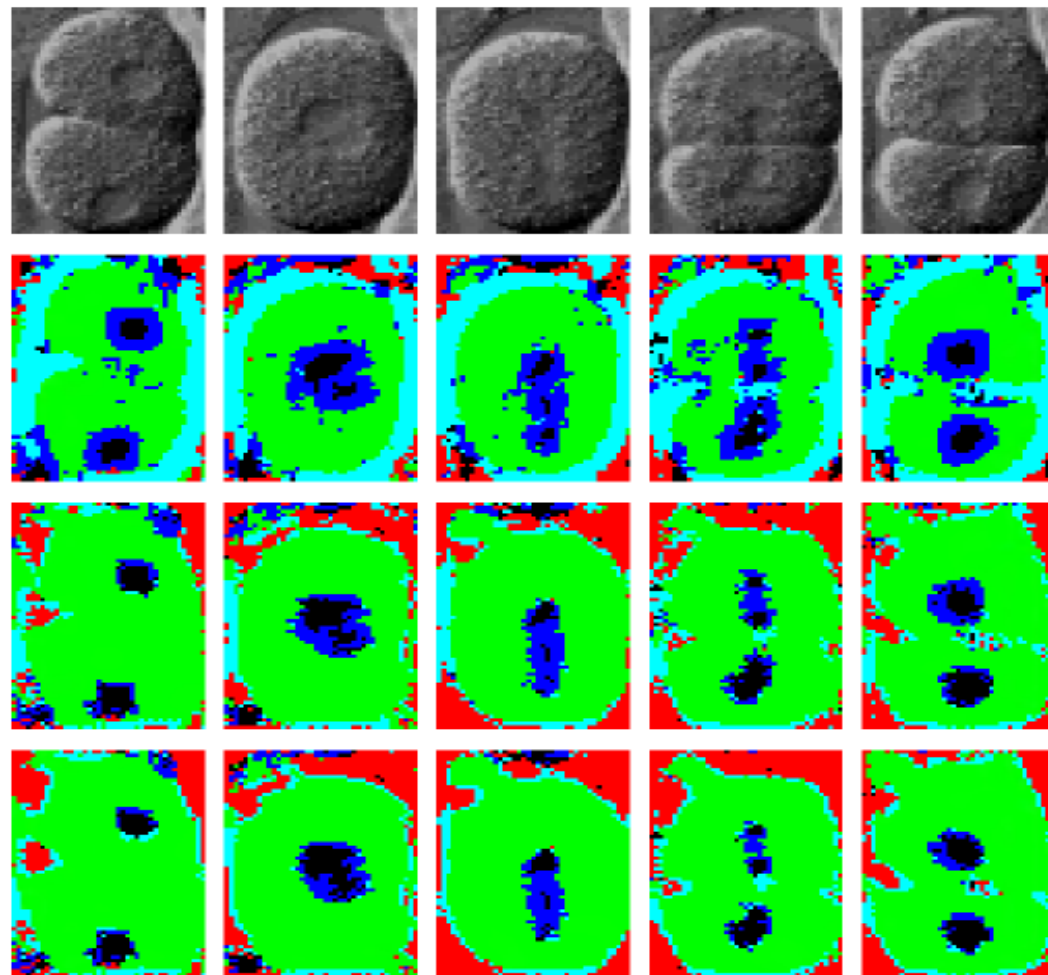
ConvNet takes a window of pixels and produces a label for the central pixel

Cleanup using a kind of conditional random field (CRF)

► Similar to a field of expert, but conditional.

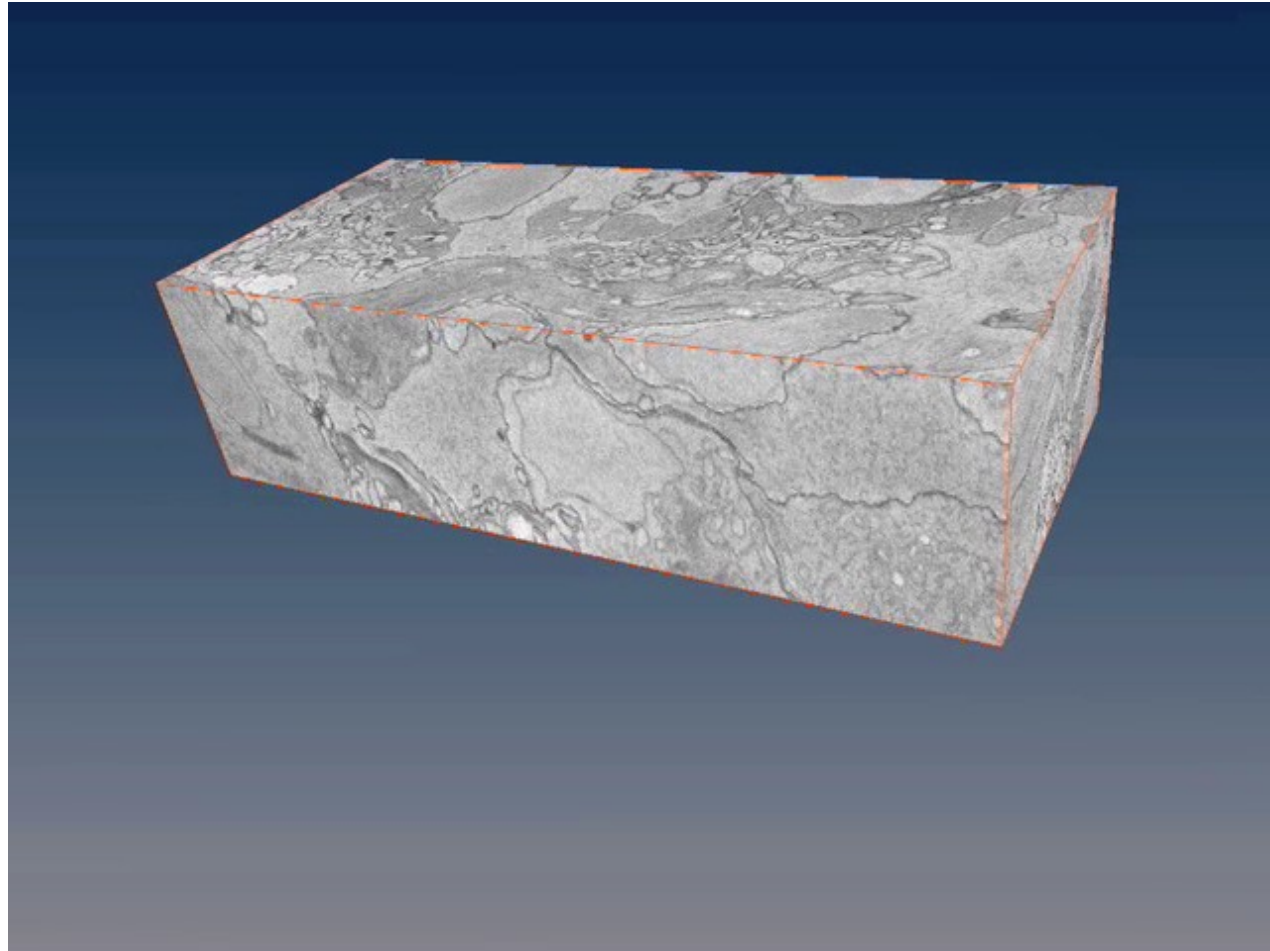
3D version for connectomics

► [Jain et al. 2007]



3D image segmentation for “connectomics”

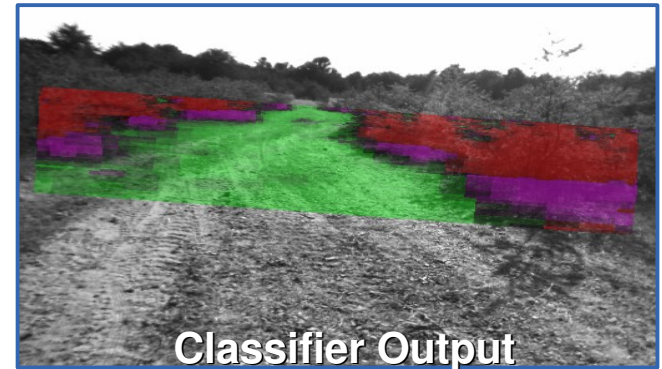
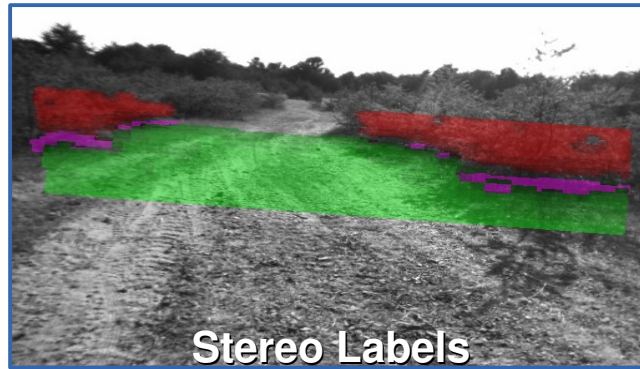
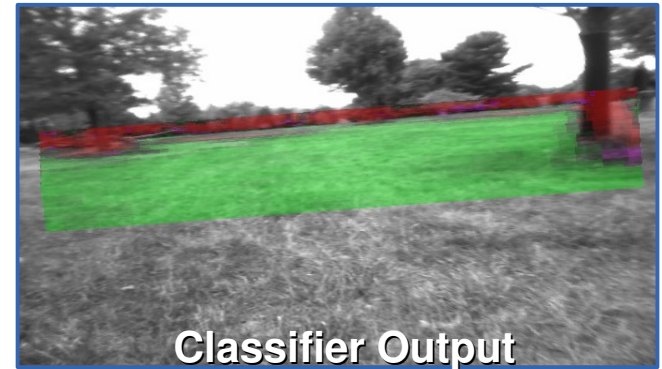
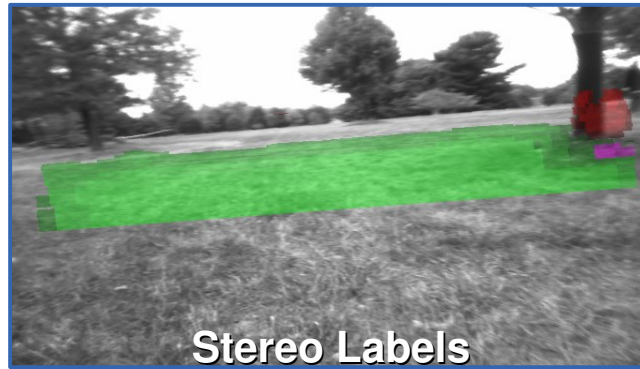
- ▶ [Jain et al. 2007]
- ▶ Volumetric images of brain tissue
- ▶ Obtained by scanning electron microscopy on brain slices a few nanometers thick
- ▶ 3D ConvNet trained to identify neuron membranes and synapses.
- ▶ Crowd-sourced data
 - ▶ <http://eyewire.org>



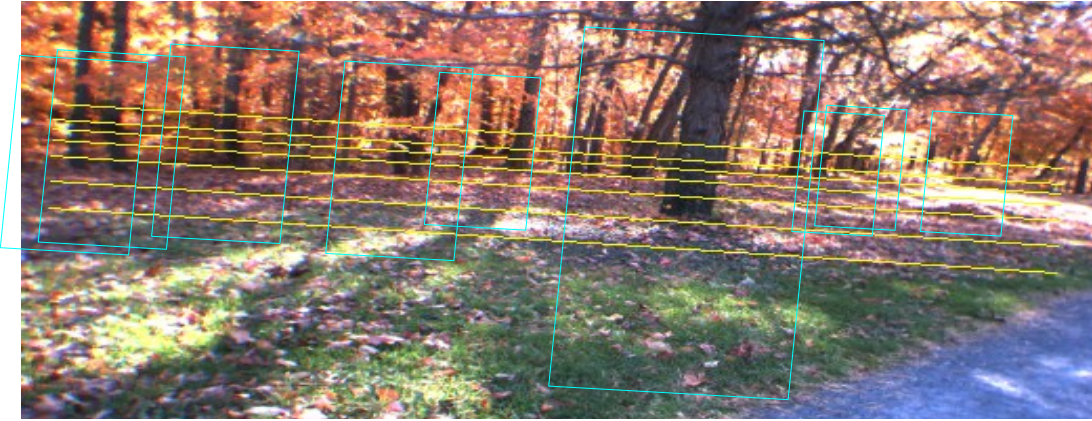
▶ “The Mind of a Mouse”, Abbott et al. Cell. Vol 182, No 6, Sep 2020. <https://doi.org/10.1016/j.cell.2020.08.010>

Semantic Segmentation [for robot navigation]

- ▶ Labeling every pixel with the category of the object it belongs to
- ▶ Here: traversable/obstacle. DARPA LAGR project [Hadsell JFR 2009]
- ▶ Training labels from stereo vision (short range)

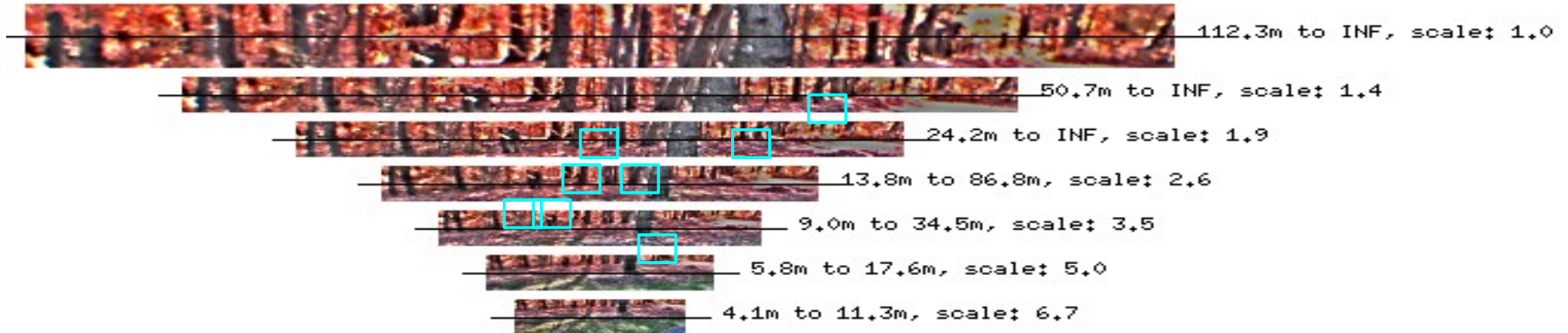


Multiscale semantic segmentation for navigation

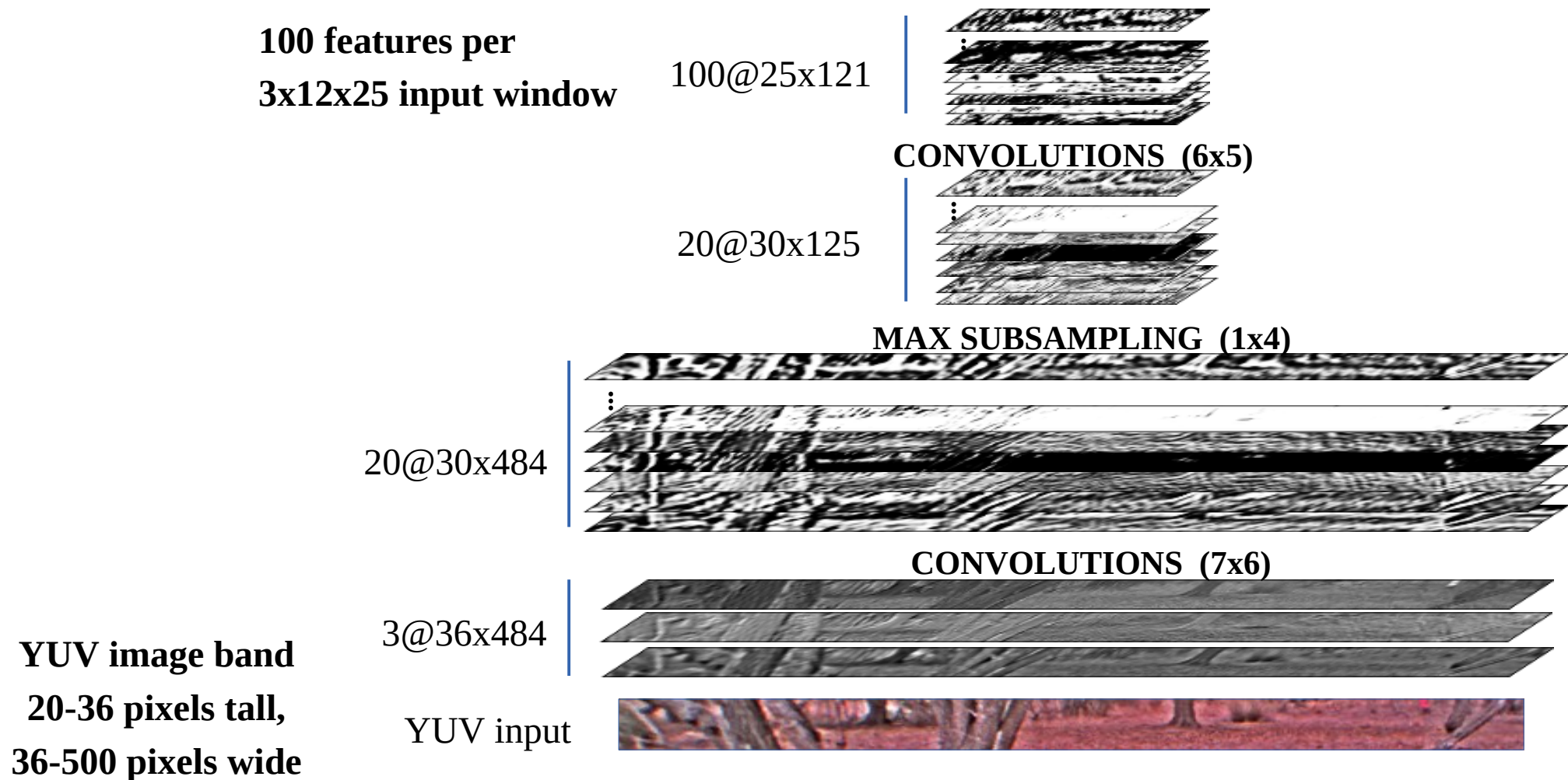


Pre-processing (125 ms)

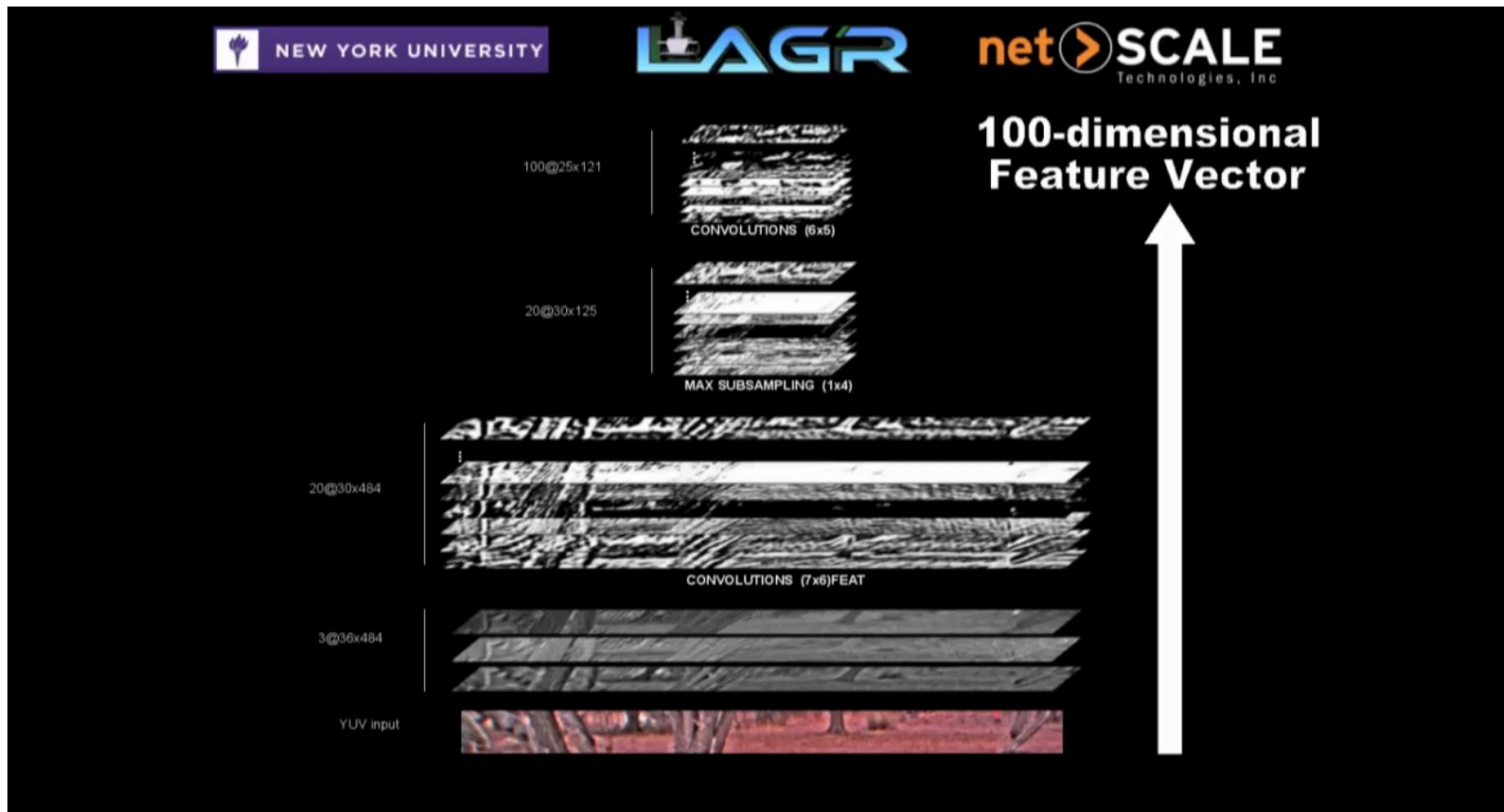
- Ground plane estimation
- Horizon leveling
- Conversion to YUV + local contrast normalization
- Scale invariant pyramid of distance-normalized image



Convolutional Net Architecture



Training a Robot to Drive Itself in Nature [Hadsell 2009]



Category-level Semantic Segmentation

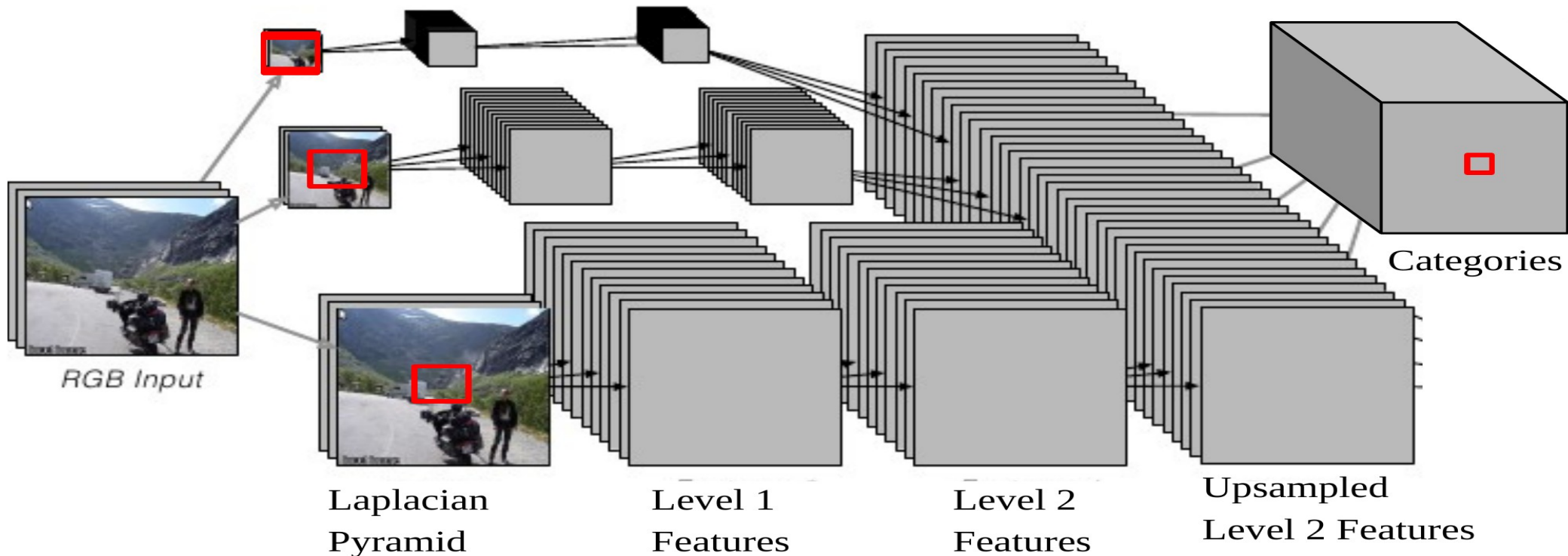


[Farabet et al. ICML 2012, PAMI 2013]

Scene Parsing/Labeling: Multiscale ConvNet Architecture

Each output sees a large input context:

- ▶ **46x46** window at full rez; **92x92** at $\frac{1}{2}$ rez; **184x184** at $\frac{1}{4}$ rez
- ▶ [7x7conv]->[2x2pool]->[7x7conv]->[2x2pool]->[7x7conv]->



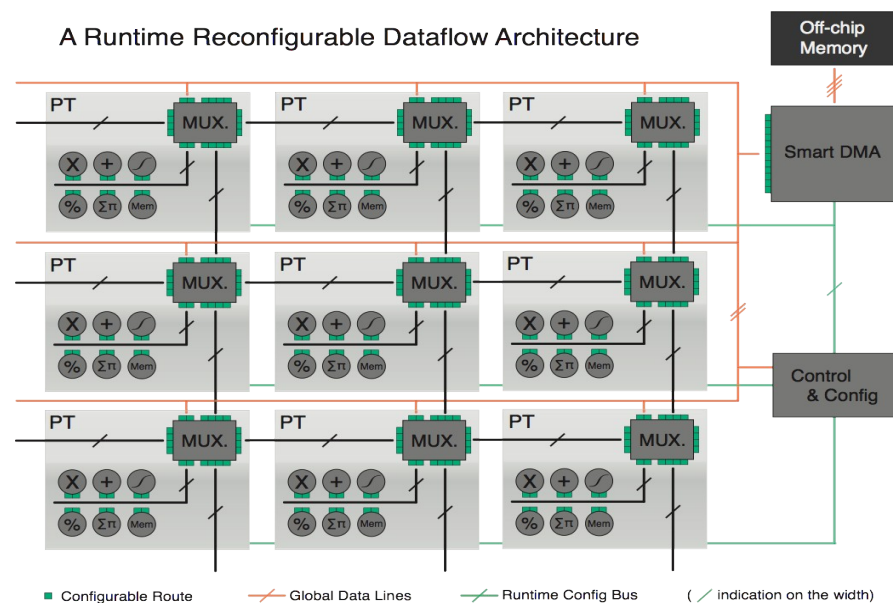
Semantic Segmentation with ConvNets [Farabet 2012]

- ▶ 33 categories
- ▶ 50ms per frame on a dedicated FPGA in 2011.

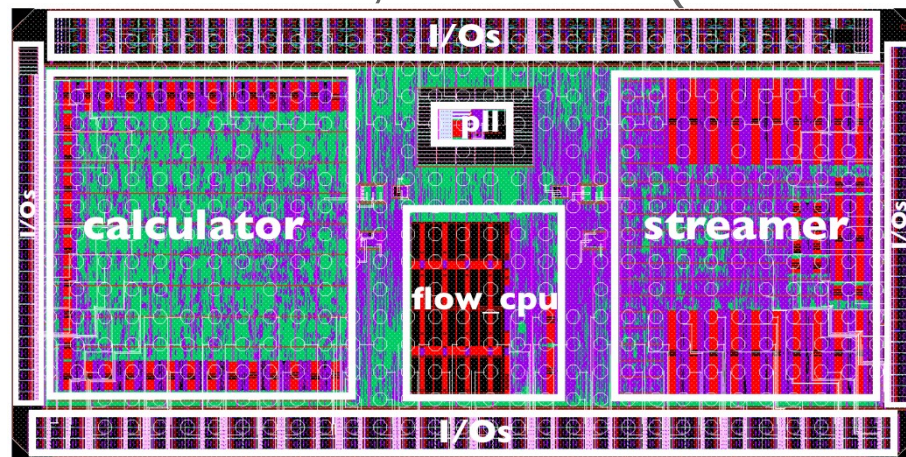


FPGA ConvNet Accelerator: NewFlow [Farabet 2011]

- ▶ **NeuFlow: Reconfigurable Dataflow architecture**
 - ▶ Implemented on Xilinx Virtex6 FPGA
 - ▶ 20 configurable tiles. 150GOPS, 10 Watts
 - ▶ Semantic Segmentation: 20 frames/sec at 320x240
 - ▶ **Exploits the structure of convolutions**

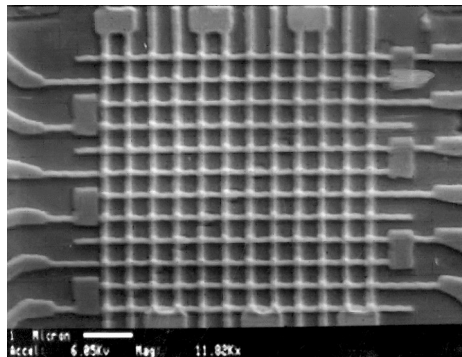


- ▶ **NeuFlow ASIC [Pham 2012]**
 - ▶ 150GOPS, 0.5 Watts (simulated)

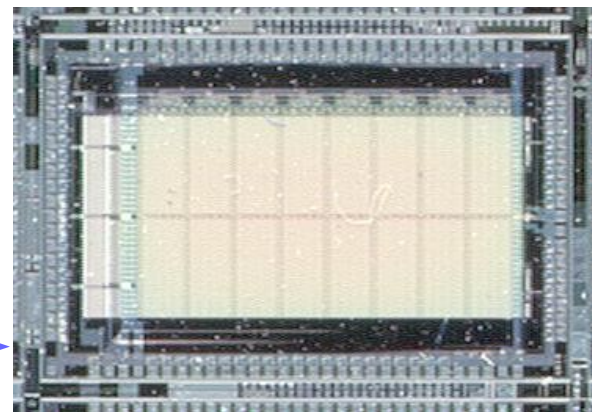
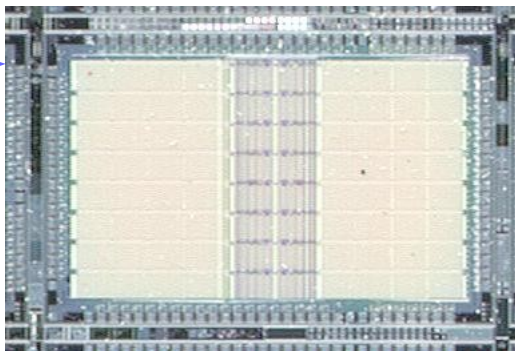
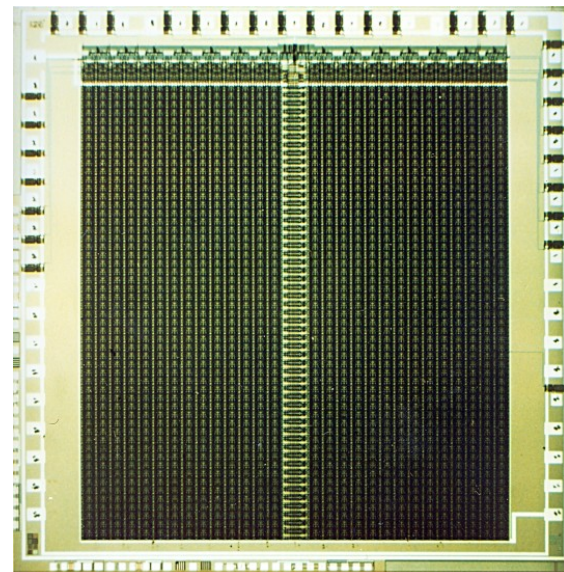


1986-1996 Neural Net Hardware at Bell Labs, Holmdel

- ▶ **1986: 12x12 resistor array** →
 - ▶ Fixed resistor values
 - ▶ E-beam lithography: 6x6microns
- ▶ **1988: 54x54 neural net**
 - ▶ Programmable ternary weights
 - ▶ On-chip amplifiers and I/O
- ▶ **1991: Net32k: 256x128 net** →
 - ▶ Programmable ternary weights
 - ▶ 320GOPS, 1-bit convolver.
- ▶ **1992: ANNA: 64x64 net**
 - ▶ ConvNet accelerator: 4GOPS
 - ▶ 6-bit weights, 3-bit activations



6 microns



The Deep Learning Revolution

Speech recognition: 2010

Image recognition: 2013

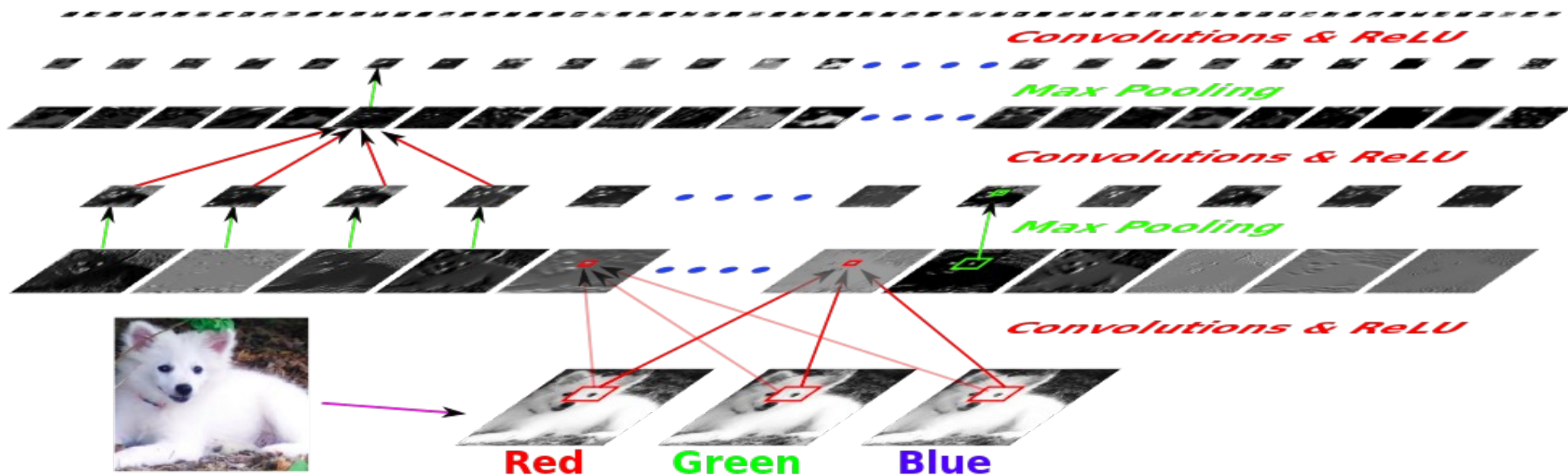
Natural language processing: 2015



Deep ConvNets for Object Recognition (on GPU)

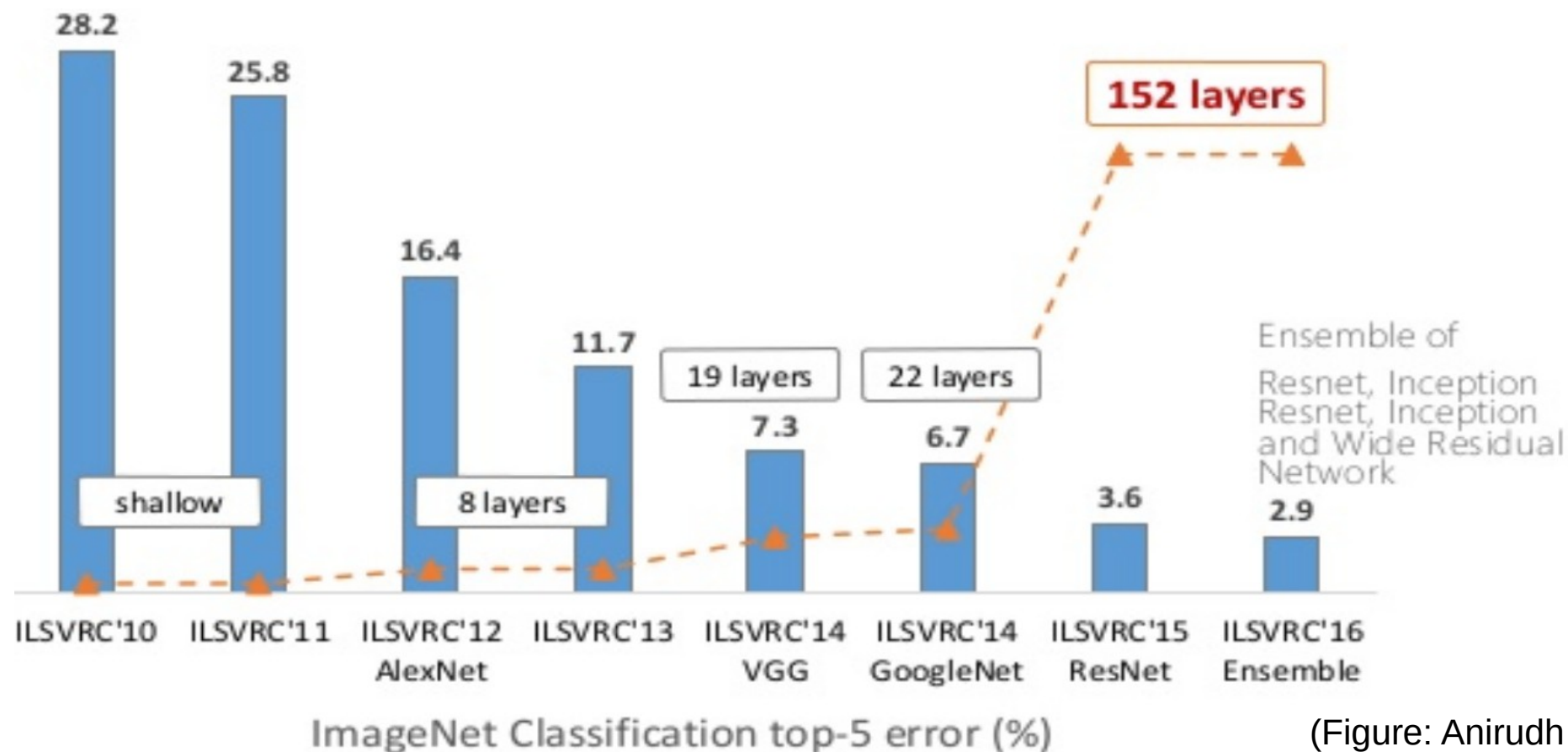
- AlexNet [Krizhevsky et al. NIPS 2012], OverFeat [Sermanet et al. 2013]
- 1 to 10 billion connections, 10 million to 1 billion parameters, 8 to 20 layers.

Samoyed (16); Papillon (5.7); Pomeranian (2.7); Arctic Fox (1.0); Eskimo Dog (0.6); White Wolf (0.4); Siberian Husky (0.4)



Error Rate on ImageNet

► Depth inflation



(Figure: Anirudh Koul)

Deep ConvNets: depth inflation!

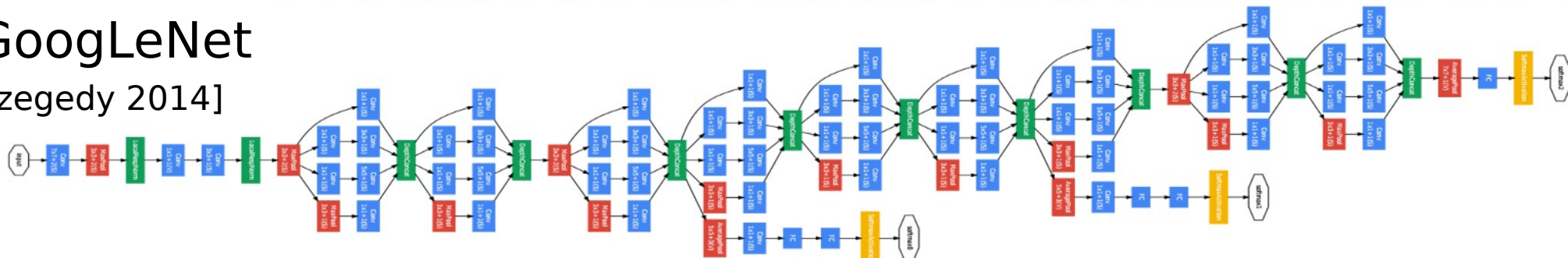
VGG

[Simonyan 2013]



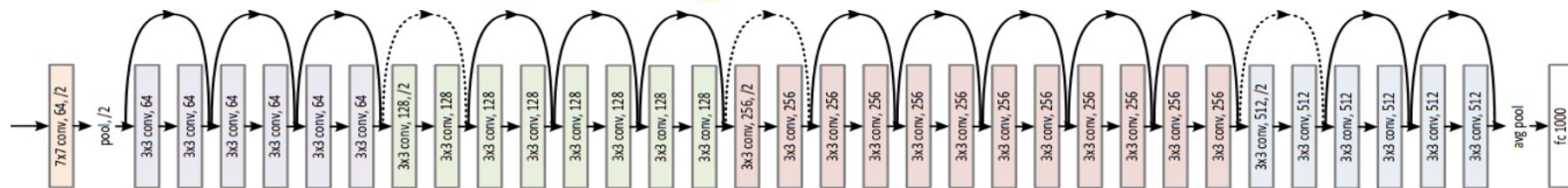
GoogLeNet

[Szegedy 2014]



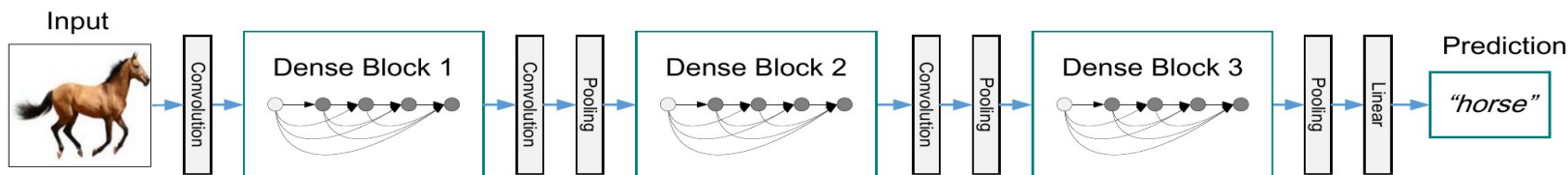
ResNet

[He et al. 2015]



DenseNet

[Huang et al 2017]



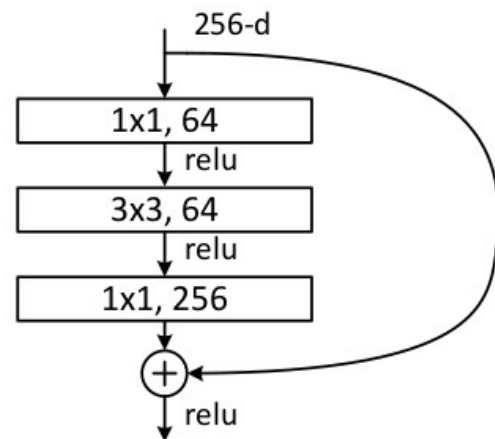
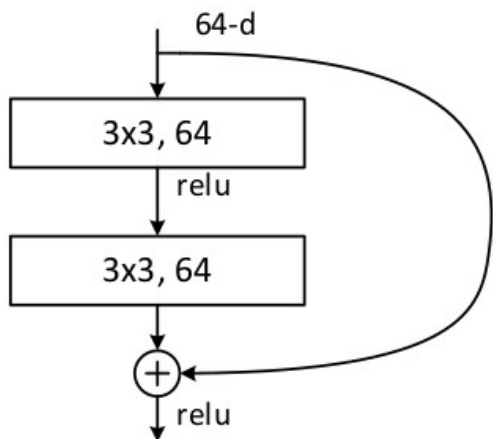
ResNet

- ▶ K He, X Zhang, S Ren, J Sun: “Deep residual learning for image recognition”, arXiv:1512.03385, (2015), Proceedings of CVPR 2016.
- ▶ 71,000 citations as of Feb 2021.
 - ▶ The most cited paper in **all of science** over the last 5 years.
 - ▶ Factoid: the 2nd most cited is “Deep Learning”, YLC, Bengio, Hinton, Nature 2015 with 35,500 citations.

- ▶ **Idea: make each layer be the identity function by default.**

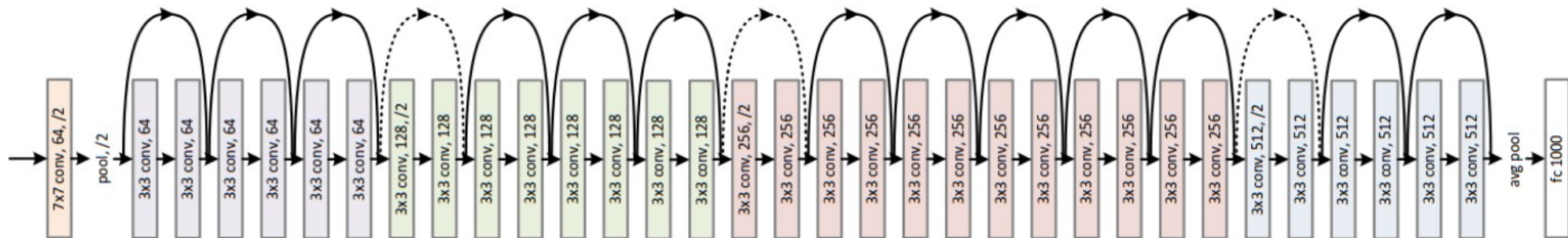
- ▶ **Neural net computes “residual” non-linear function.**

- ▶ **Good with many layers**



Example: ResNet-34, ResNet-50

- ▶ **ResNet-50 is the workhorse of modern image recognition.**
 - ▶ Standard implementation in PyTorch
- ▶ **Many detection systems use a so-called “standard ResNet-50 backbone”**
- ▶ **Many variants since 2015: ResNext, DenseNet, EfficientNet.....**
- ▶ **Leaderboard of various architectures on ImageNet:**
 - ▶ <https://paperswithcode.com/sota/image-classification-on-imagenet>



GOPS vs Accuracy on ImageNet vs #Parameters

► [Canziani 2016]

► ResNet50 and ResNet100 are used routinely in production.

► Each of the few billions photos uploaded on Facebook every day goes through a handful of ConvNets within 2 seconds.

